Marked geometry of strict Conway compositions

Abstract

An ordered expression in crossing-circle tangles, products and physical turns determines a link diagram and a real wire polytope. We construct the complete hyperbolic metric whenever that polytope has a strict state. Four separate boundary-edge labels allow the pyramid-and-prism construction to pass through turned elementary children and arbitrary strict closing pairs. We prove convexity, retain every original shaded triangle, and identify the closed assembly with the literal projection cut of the specified link. Whole complex face maps then give its marked deformation, with parabolic main cusps. The original primitive filling equations have a unique small solution for independently long signed slopes. Smooth completion and an absolute fundamental-chain comparison identify the volume with a critical value of a classical cell potential. No asymptotic formula for a quantum invariant is used or asserted.

1 The fixed diagram and the geometric statements

A crossing circle is an auxiliary unknotted component bounding a disk punctured twice by the main strands. We retain its disk and its original framing. Let h be the physical quarter-turn of this elementary two-strand tangle, with full twists removed. A construction is a finite expression in distinct copies of h, ordered products and physical clockwise quarter-turns T. No reduced algebraic expression replaces the original diagram in this paper.

The four endpoints of a box are numbered $0, 1, 2, 3$ cyclically, starting at the bottom left. In the stack FG, with F below G, join F_2 to G_1 and F_3 to G_0. Its exposed tuple is (F_0, F_1, G_2, G_3). A turn replaces a tuple by (C_1, C_2, C_3, C_0) and moves the entire box and all its disks. Cylindrical closure C joins endpoints 0 to 3 and 1 to 2. Our closed diagrams are $C(FG)$ with arbitrary such F, G subject to the following condition.

Definition 1.1. A strict wire state assigns a leaf variable $a \in (0, 1/2)$ and an output variable to every open subexpression. A turn complements its variable. Every product retained below the closing root satisfies

$$\max(x, y) < z < x + y < 1. \tag{1.1}$$

The two closing outputs are $b, 1 - b$. The final forced gate is removed, but this complementary relation is retained. An expression is strict if the resulting real polytope has a strict state.

Thus strictness is a condition on this particular presentation. A turned leaf or a bare leaf may occur at a closing child. This is larger syntax than the completed/completed closing grammar of [2]; its global metric theorem will not be applied here.

Fix $\varepsilon_i \in \{-1, 0, 1\}$ at the H original leaves. It specifies the retained signed half twist; $\varepsilon_i = 0$ is the pure disk pairing. Let Y_E^{ε} be the exterior in S^3 of the resulting augmented link, retaining all H crossing circles and every main component. The source-independent functions

used below are

$$J(u) = \int_0^u \log(2\sin \pi t)\,\mathrm{d}t \quad (0 \le u \le 1), \qquad J(1-u) = -J(u), \tag{1.2}$$

$$g(a) = J(2a) - 4J(a), \tag{1.3}$$

$$\begin{aligned} m(x,y,z) = {} & J(x+y) + J(z) - J(x) - J(y) \\ & - J(z-x) - J(z-y) - J(x+y-z). \end{aligned} \tag{1.4}$$

Let $\mathcal{F}$ be the sum of g over leaves and m over retained products, and let F_0 denote its maximum. The variational theorem of [1] supplies the unique strict optimizer and its negative definite Hessian in independent wires. We state the precise input in Section 2; its proof does not use a global metric or a quantum noncancellation theorem.

Theorem 1.2 (Complete marked exterior)**.** *For every fixed strict expression and retained half-twist pattern, Y_E^ε has a complete finite-volume hyperbolic metric assembled from two copies of a convex right-angled ideal polyhedron. Its volume is*

$$\mathrm{Vol}(Y_E^\varepsilon) = 2\pi F_0. \tag{1.5}$$

At each crossing circle the original meridian μ_i and disk longitude λ_i are a primitive ordered basis. Their complete longitude-to-meridian translation ratio is

$$\tau_i = \frac{2\mathrm{i}c_i}{1 - \varepsilon_i \mathrm{i}c_i}, \qquad c_i = \cot(\pi a_i) > 0, \qquad \operatorname{Im}\tau_i > 0. \tag{1.6}$$

All original main components have complete cusp ends.

The primitive basis fixes the filling convention. For a subset $I \subseteq \{1,\dots,H\}$ and integers p_i, fill precisely the auxiliary slopes $\mu_i + p_i\lambda_i$, $i \in I$. Denote the original filled exterior by $Y_E^\varepsilon(\mathbf{p}; I)$. Cutting the original disk, rotating through p_i full turns and removing its crossing circle inserts the physical power $\varepsilon_i - 2p_i$ in that same disk frame. The disjoint disk neighborhoods show directly that the ambient manifold is still S^3. We never fill a main cusp.

Theorem 1.3 (Primitive signed fillings)**.** *There is a holomorphic classical potential $\Phi_\varepsilon(\boldsymbol{\chi})$, with $\Phi_\varepsilon(0) = F_0$, and a framed representation of the original group near its oriented complete holonomy, whose primitive similarity logarithms are*

$$U_i = -2\pi\mathrm{i}\chi_i, \qquad V_i = -2\partial_i\Phi_\varepsilon. \tag{1.7}$$

Every main peripheral group remains parabolic. Set $\chi_i = 0$ outside I. There is a bound $B_{E,\varepsilon,I}$ such that, whenever all $|p_i| > B_{E,\varepsilon,I}$, the restricted action

$$S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}) = \Phi_\varepsilon(\boldsymbol{\chi}) + \frac{\pi\mathrm{i}}{2}\sum_{i\in I}\frac{(\chi_i - 1)^2}{p_i} \tag{1.8}$$

has a unique critical point in a fixed neighborhood of zero. The representation there completes to a smooth complete finite-volume hyperbolic metric on the original $Y_E^\varepsilon(\mathbf{p}; I)$, and

$$\mathrm{Vol}(Y_E^\varepsilon(\mathbf{p}; I)) = 2\pi \operatorname{Re} S_{\varepsilon,\mathbf{p}}(\boldsymbol{\chi}_*). \tag{1.9}$$

The signs and relative sizes of the filling integers are independent. If I contains all auxiliaries and the literal strand pairing has one component, the right side is the volume of that knot complement.

The uniqueness assertion is local in this prescribed family, not a classification of the character variety. The potential is defined from the cells in Section 6; no selected quantum action is an input. A maximum over the finitely many ε, I gives one bound for a fixed strict expression if desired. An empty filling subset recovers Theorem 1.2.

The classical geometry of fully augmented links and their half-twist partners is well established; see [6]. Dehn filling and holomorphic peripheral potentials belong to the framework of [7]. The work here is the ordered individual-edge assembly and its marked continuation for strict expressions, with the original manifold and the absolute normalization proved explicitly.

2 Variational input and local cells

We use the following specific conclusions of the strict-profile companion [1, Lemma 5.2, Theorem 6.4, Lemma 7.1 and Proposition 7.2]. For each nonempty strict fixed-output domain, the maximum over the internal wires is strict, unique and nondegenerate. Its profile $f_E(b)$ is real analytic on that domain. The analogous closed statement holds on Definition 1.1's reduced affine constraint space. The exact open types and product table are

$$L = (0, 1/2), \quad H = (1/2, 1), \quad A = (0, 1), \qquad \begin{array}{c|ccc} & L & H & A \\ \hline L & A & H & A \\ H & H & \varnothing & H \\ A & A & H & A \end{array}. \tag{2.1}$$

Every point of each indicated interval is attained. A leaf has type L; a turn exchanges L, H and preserves A. The nonempty closing pairs are AA, AL, AH, LH and their reversals.

For an optimized output define

$$B = \cot \pi b, \quad \varkappa = e^{-f_E'(b)}, \qquad X^2 = \frac{1 + B^2}{1 + \varkappa}, \quad Y^2 = \varkappa X^2, \quad X, Y > 0. \tag{2.2}$$

Then $X^2 + Y^2 - B^2 = 1$, and the imported profile identities give

$$L : (B, X, Y) = (B, B, 1), \qquad H : (B, X, Y) = (-Y, 1, Y), \qquad A : -Y < B < X. \tag{2.3}$$

A turn sends (B, X, Y) to $(-B, Y, X)$. At a stationary product the two children have a common X, and, writing $S = B_1 + B_2 > 0$,

$$B_o = \frac{B_1 B_2 + X^2 - 1}{S}, \qquad Y_o = \frac{Y_1 Y_2}{S}. \tag{2.4}$$

These are scalar variational identities; they do not identify an open quadrilateral with a physical four-punctured tangle boundary.

2.1 The ideal-tetrahedron normalization

We use upper half-space with ideal boundary $\mathbb{C} \cup \{\infty\}$. A geodesic plane is vertical over a line or is a hemisphere orthogonal to the boundary plane. A horizontal horosphere at ∞ has a Euclidean induced metric up to scale. For positive a, b, c with $a + b + c = 1$, the classical volume of an ideal tetrahedron with dihedral angles $\pi a, \pi b, \pi c$ is

$$\begin{aligned} V(a, b, c) &= \Lambda(\pi a) + \Lambda(\pi b) + \Lambda(\pi c) = -\pi\{J(a) + J(b) + J(c)\}, \\ \Lambda(t) &= -\int_0^t \log(2 \sin u)\, \mathrm{d}u. \end{aligned} \tag{2.5}$$

We import this normalized local formula, including its continuous zero value at a flat tetrahedron, from classical hyperbolic geometry [5]. All cell volumes below are ordinary sums of positive tetrahedron volumes.

2.2 The leaf pyramid

For $0 < a < 1/2$, put $E = e^{2\pi i a}$. The convex hull of

$$O = \infty, \qquad (C_0, C_1, C_2, C_3) = (1, E, -1, -E) \tag{2.6}$$

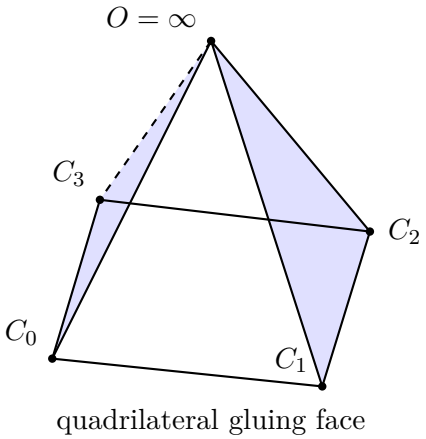
is an ideal quadrangular pyramid. Its four finite vertices form a rectangle. Its side planes are vertical over the four rectangle edges; its base plane is the unit hemisphere. These five supporting planes enclose exactly the convex hull. Adjacent vertical planes meet at right angles, so every apex edge has angle $\pi/2$.

Call base edges $01, 23$ free and $12, 30$ cap edges. Intersecting a vertical chord plane with the unit hemisphere gives interior angle equal to half the angle subtended by that chord at the circle center. Thus free edges have angle πa, and cap edges have angle $\pi(1/2 - a)$. Color (O, C_1, C_2) and (O, C_3, C_0) shaded and the other two side triangles white. The shaded faces will retain the two sides of the original half crossing disk. The ordered quadrilateral base is a gluing face in the cut construction.

The diagonal C_0C_2 divides the pyramid into two positive tetrahedra with angle triple $(a, 1/2 - a, 1/2)$. By (2.5),

$$\mathrm{Vol}(\text{leaf}) = 2V(a, 1/2 - a, 1/2) = \pi g(a).$$

The last equality follows either by differentiating the two expressions and evaluating at $a = 0$, or from $J(2a) = 2J(a) - 2J(1/2 - a)$.



Ideal hyperbolic pyramid.

All five vertices are ideal. The shaded triangular faces are (O, C_1, C_2) and (O, C_3, C_0).

In the boundary plane:
$C_0 = 1$, $C_1 = e^{2\pi i a}$,
$C_2 = -1$, $C_3 = -e^{2\pi i a}$.

Figure 1: The ideal hyperbolic leaf pyramid. The two shaded triangles record the sides of a half crossing disk. The ordered quadrilateral is a gluing face of the cut cells; it is not identified with the physical four-punctured boundary of a tangle box.

2.3 The multiplication prism

Given a strict product, put $p = z - x$, $q = z - y$, $r = x + y - z$, $w = 1 - x - y$. All four numbers are positive, and $p + q + 2r + w = 1$. Choose four points on the unit circle:

$$v_0 = 1, \qquad v_1 = e^{2\pi \mathrm{i} p}, \qquad v_2 = e^{2\pi \mathrm{i}(p+r)}, \qquad v_3 = e^{2\pi \mathrm{i}(p+r+q)}. \tag{2.7}$$

Their successive half-arcs are $p, r, q, r + w$. The extended chords v_0v_1 and v_3v_2 meet at a point P. The triangle (P, v_1, v_2) has angles $\pi w, \pi y, \pi x$ at those respective vertices. The sine

rule, with chord lengths $2\sin\pi p$, $2\sin\pi q$, $2\sin\pi r$, gives

$$\begin{aligned} P &= v_1 + \frac{\sin\pi x \sin\pi r}{\sin\pi w \sin\pi p}(v_1 - v_0) \\ &= v_2 + \frac{\sin\pi y \sin\pi r}{\sin\pi w \sin\pi q}(v_2 - v_3). \end{aligned} \tag{2.8}$$

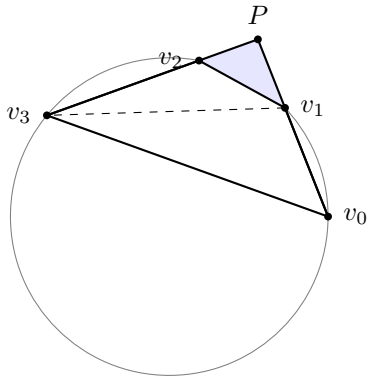
Both coefficients are positive. Thus the orders on the two lines are v_0, v_1, P and v_3, v_2, P.

Attach the tetrahedron (∞, P, v_1, v_2) to the pyramid $(\infty, v_0, v_1, v_2, v_3)$ across (∞, v_1, v_2). Their interiors lie on opposite sides of that vertical plane. The faces over v_0v_1 and v_1P are coplanar, as are those over v_3v_2 and v_2P. Merging them leaves three quadrilateral faces and two triangular faces:

$$F_x = (\infty, v_0, v_1, P), \qquad F_y = (\infty, v_3, v_2, P), \qquad F_z = (v_0, v_1, v_2, v_3),$$
$$(P, v_1, v_2), \qquad (\infty, v_0, v_3).$$

The last two faces are white.

There are five supporting planes: the vertical planes over v_0v_1, v_2v_3, v_3v_0, and the hemispheres over the circles (v_0, v_1, v_2, v_3) and (P, v_1, v_2). The power of P with respect to the unit circle is positive, because P is beyond v_1 on the chord v_0v_1. Likewise the powers of v_0, v_3 with respect to the circle (P, v_1, v_2) are positive: along its two secants the points have orders v_0, v_1, P and v_3, v_2, P. Consequently both attached pieces are in the same inward half-space of each of these five planes. Their union is the convex ideal triangular prism bounded by those faces. This proves support and nonintersection throughout the strict domain, rather than for one drawn example.



Boundary-plane construction.
$P = (v_0v_1) \cap (v_2v_3)$ lies outside the unit circle.
The sixth ideal vertex is ∞.
The three quadrilateral gluing faces have ordered vertices (∞, v_0, v_1, P), (∞, v_3, v_2, P) and (v_0, v_1, v_2, v_3).

$p = .12, \quad r = .10, \quad q = .17, \quad w = .51$

Figure 2: Boundary-plane support data for the multiplication prism; its sixth ideal vertex is ∞. The point P is the intersection of the extended chords, with the two orders proved in (2.8). The circle through P, v_1, v_2 and the unit circle support the two bounded faces.

The original lower and upper attachments and the output order are

$$F : C \mapsto (v_0, v_1, P, \infty), \quad G : C \mapsto (\infty, P, v_2, v_3), \quad C^o = (v_0, v_1, v_2, v_3). \tag{2.9}$$

Reading angles between the supporting planes gives

edge	$Pv_1, \infty v_0$	$Pv_2, \infty v_3$	v_1v_2, v_0v_3	$P\infty$	$v_0v_1,\ v_2v_3$
angle/π	x	y	$1-z$	w	$p,\ q$

. (2.10)

For a circle and a chord plane, the dihedral angle is the opposite inscribed angle. Applied to (P, v_1, v_2) this gives x, y at Pv_1, Pv_2; the angle between the two vertical chord planes is πw. At the unit-circle base the half-arcs give p, q. The remaining angles follow at the ideal vertices: their Euclidean links are triangles, so the interior dihedral angles sum to π. For example $x + y + w = 1$ at P, and $x + p + (1 - z) = 1$ at v_1. This verifies every row of the table.

Divide the original quadrangular pyramid across v_0v_2. Its tetrahedron triples are $(p, r, 1 - p - r)$ and $(p + r, q, r + w)$. In their volumes the terms at $p + r$ and $1 - p - r$ cancel. The attached tetrahedron has triple (y, x, w). Thus the sum is also

$$\begin{aligned} \mathrm{Vol}(\text{prism}) &= V(y, x, w) + V(p, q, 2r + w) + V(p + q, r, r + w) \\ &= \pi m(x, y, z). \end{aligned} \tag{2.11}$$

To check the first rewriting, the two extra terms $\Lambda(\pi(p+q))$ and $\Lambda(\pi(2r+w))$ cancel because their arguments sum to π. For the second equality use $x = q + r$, $y = p + r$, $z = p + q + r$, $r + w = 1 - z$ and $w = 1 - x - y$ in (2.5). All triples used here are positive and sum to one.

2.4 Matching the whole ordered face

For a cyclically ordered ideal quadruple define

$$\varkappa(a, b, c, d) = \frac{(a - b)(c - d)}{(a - d)(b - c)} > 0 \tag{2.12}$$

with the usual limits at infinity. Sending three vertices to $(\infty, 0, 1)$ shows that this positive number determines the entire ordered ideal quadrilateral, including its shear. A one-step cyclic shift and a reversal fixing the first vertex invert $\varkappa$; full reversal preserves it.

Formula (2.8) immediately gives the first two ratios below as directed ratios on the chord lines; the third follows from the four unit-circle chord lengths:

$$\varkappa(F_x) = \frac{\sin \pi x \sin \pi r}{\sin \pi w \sin \pi p} = e^{-m_x}, \qquad \varkappa(F_y) = \frac{\sin \pi y \sin \pi r}{\sin \pi w \sin \pi q} = e^{-m_y}, \tag{2.13}$$

$$\varkappa(F_z) = \frac{\sin \pi p \sin \pi q}{\sin \pi z \sin \pi r} = e^{-m_z}. \tag{2.14}$$

The derivative equalities follow by differentiating the seven J terms in (1.4); for example $m_x = \log(\sin \pi w \sin \pi p / (\sin \pi x \sin \pi r))$. The leaf ratio is $\tan^2(\pi a) = e^{-g'(a)}$.

The lower tuple in (2.9) is a cyclic shift of F_x and the upper is a reversal fixing the first vertex of F_y, so both invert the input ratios. The stationary equations $f'_F(x) + m_x = 0$ and $f'_G(y) + m_y = 0$ therefore match all four ordered vertices of each attachment. The output derivative is m_z, which proves $\varkappa = e^{-f'_E}$ inductively. A physical turn cyclically relabels the quadruple and replaces b by $1 - b$, again inverting the ratio. Finally $C^F_j \mapsto C^G_{3-j}$ is full reversal. The closing equation $f'_F(b) = f'_G(1 - b)$ therefore matches the whole closing face. None of these steps replaces a complex shape by its absolute value.

3 Four boundary labels and convex assembly

An open cell has one ordered quadrilateral gluing face $Q = (C_0, C_1, C_2, C_3)$. Number its edges $j = (j, j + 1)$ modulo four. Edges $0, 2$ are called free, and $1, 3$ cap edges. These names record the original leaf order, not an unmarked symmetry of the whole tangle. Set $\eta_j = 1$ if the face beside edge j, other than Q, is shaded, and set $\eta_j = 0$ if it is white. The four bits $\boldsymbol{\eta} = (\eta_0, \eta_1, \eta_2, \eta_3)$ will be called the boundary mask.

Proposition 3.1. *At every open subexpression evaluated at its strict stationary wires there is a convex ideal cell with a maximal, nondegenerate ordered quadrilateral output. Its boundary-edge angles are*

$$\frac{\theta_j}{\pi} = \begin{cases} b - \eta_j/2, & j = 0, 2, \\ 1 - b - \eta_j/2, & j = 1, 3. \end{cases} \tag{3.1}$$

Every other surviving edge is right angled and separates a white face from a shaded triangle. The possible supports satisfy

$$L : \eta_0 = \eta_2 = 0, \qquad H : \eta_1 = \eta_3 = 0, \qquad A : \boldsymbol{\eta} = 0. \tag{3.2}$$

Every original shaded triangle remains an entire face. The four output corners are four active original terminal singletons. Every other main ideal vertex is a permanently consumed pair of terminals; crossing-circle apices stay separate.

The statement permits a type H output with only one shaded free edge. Replacing its two free angles by a common angle would therefore lose information essential to the next gluing.

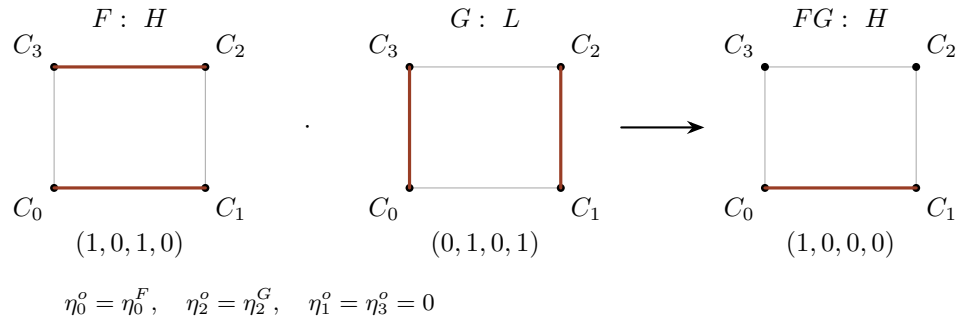


Figure 3: Individual boundary labels in an HL product. A colored edge has a shaded adjacent face; the other edges have a white adjacent face. The turned leaf has mask $(1, 0, 1, 0)$, the upper leaf has mask $(0, 1, 0, 1)$, and the product has mask $(1, 0, 0, 0)$. Its two free angles are $\pi(z - 1/2)$ and πz. The pictures display the ordered quadrilateral gluing faces.

3.1 Every seam angle

For a leaf the mask is $(0, 1, 0, 1)$, so (3.1) is the pyramid angle calculation. A turn rotates the mask by one and replaces b by $1 - b$. This proves the assertion under turns and exchanges the support conditions for L, H.

At a product the output mask is exactly

$$\boldsymbol{\eta}^o = (\eta_0^F, 0, \eta_2^G, 0). \tag{3.3}$$

Indeed the two output cap sides adjoin the prism's white triangles, and each output free side inherits its indicated child side. A child cap seam has angle $\pi(1 - \eta_j/2)$: a zero bit merges two white pieces flatly, while a one bit leaves a right shaded–white edge. At the edge common to both prism inputs, the total angle is

$$\pi \left[w + x - \frac{\eta_2^F}{2} + y - \frac{\eta_0^G}{2} \right] = \pi \left[1 - \frac{\eta_2^F + \eta_0^G}{2} \right]. \tag{3.4}$$

Both bits could be one only with two type H inputs, forbidden by (2.1). Thus this seam is either flat white–white or right shaded–white. The remaining free output angles are $\pi(p + x - \eta_0^F/2)$ and $\pi(q + y - \eta_2^G/2)$; the cap angles are $\pi(1 - z)$. This gives (3.1).

The support assertion also follows case by case from (2.1). An output of type A has no type H child, so both inherited free bits vanish. An output of type H has zero cap bits by (3.3). Turns give all type L cases. We assert these support inclusions, not equality with the two-bit leaf masks.

For successive convex attachments one must check the intermediate stage, not only the final totals. Attach F to the prism first. Its free seams have angles

$$\pi(z - \eta_0^F/2), \qquad \pi(1 - y - \eta_2^F/2).$$

If the first bit is one, F has type H, hence $z > x > 1/2$. If the second bit is one, $x > 1/2$ and $x + y < 1$ give $y < 1/2$. Both displayed angles are therefore positive and strictly less than π. In particular the other prism input remains a supported nondegenerate quadrilateral. The cap seams have already been checked. The second attachment is now justified by (3.4).

3.2 Convexity includes the ideal boundary

Here is the geometric argument used at each attachment. Place two convex finite-faced ideal polyhedra on opposite sides of the plane of the matched quadrilateral. Their finite intersection is that face. The finite union is closed, connected and locally convex: the only new boundary points lie on the seam edges, where the total angle is at most π. There are no finite seam vertices.

Two finite points in the union have a shortest path in it. Indeed a length-bounded minimizing sequence stays in a compact ambient hyperbolic ball. Constant-speed parametrization and Arzelà–Ascoli give a limiting path; closedness keeps it in the union. Local convexity makes each small subarc the ambient geodesic. The complete path is therefore the unique ambient geodesic segment. This proves convexity of the union.

There is a further ideal-point issue. The two ideal closures can meet only on the boundary circle of the common supporting plane. The intersection of either cell with that plane is exactly its maximal quadrilateral face. An additional ideal vertex on the circle would enlarge that face past one of its edges, making the edge flat; all four supporting angles there are strictly nonflat. Equivalently, in the Klein model the closed quadrilateral meets the sphere at infinity only in its four vertices. Thus the only ideal coincidences are the four specified corner identifications. This argument applies again to the second input, which the intermediate angle check kept strictly supported.

Every output angle in (3.1) lies in $(0, \pi)$ on its actual type interval. Hence the output remains a maximal supported quadrilateral and the induction continues. Every shaded-triangle seam is right, never flat. Only white faces merge: no shaded edge or triangle is deleted, split or merged with another shaded face. A leaf apex is never in an attaching quadrilateral. It retains its four right edges and its two original shaded sectors.

3.3 Terminal pairs never become chains

Give leaf i the four distinct terminal symbols $(i, 0), \dots, (i, 3)$ and the distinct apex symbol O_i. The purely literal endpoint rules prove the following invariant independently of the metric. An open expression has four active singleton terminals; all other terminals form disjoint consumed two-element pairs. A turn only permutes the active tuple. Product consumes exactly

$$F_2 \sim G_1, \qquad F_3 \sim G_0, \tag{3.5}$$

and keeps F_0, F_1, G_2, G_3 active. The child leaf sets are disjoint. A consumed pair is never subsequently attached or exposed. Thus no class grows to three terminals. The geometric

maps (2.9) make exactly these pairs; the ideal-point argument excludes any additional identification.

At an exposed corner the two adjacent mask bits have sum at most one. If their sum is zero, its cyclic faces are the gluing quadrilateral, white, shaded, white; if the sum is one there are only the quadrilateral, white and shaded faces in the retained order. This also follows by taking the ideal Euclidean link: the two quadrilateral-edge angles sum to π or $\pi/2$, and the remaining edge angles are right. Each exposed corner therefore carries exactly one original shaded sector. At (3.5), the two sectors concatenate in the original ribbon order. This proves every assertion of Proposition 3.1.

4 The closed polyhedron and the original link

At the closing root glue the entire face by

$$C_j^F \longmapsto C_{3-j}^G. \tag{4.1}$$

The profile equation already matches its ordered metric. An edge j on the first child matches edge $2-j$ on the second. Its total angle is π minus half the sum of the two mask bits, in units of π. Two shaded free bits would require an HH closing pair; two shaded cap bits would require LL. Neither is strict. The convex-union proof therefore applies, and all remaining edges of the closed polyhedron P_E are right shaded–white edges.

The four closing pairs in (4.1) are between distinct root-child leaf sets. Together with (3.5) they pair every original terminal once, never with a terminal of the same leaf. Hence each leaf still has four distinct corner vertices. Its two shaded triangles are distinct and meet only at their original apex. This rules out a transitive corner collapse at closure.

The boundary of the actual convex polyhedron is a sphere. Its white faces are maximal convex ideal polygons. They cannot be bigons or zero-width regions. No two shaded triangles share an edge: every surviving edge has a white side. These observations will matter for both the topological cut and the cusp returns.

4.1 Recovering every colored face from the literal wires

Compile the same boundary without prisms. At leaf i put the two shaded triangles

$$(O_i, C_{i1}, C_{i2}), \qquad (O_i, C_{i3}, C_{i0})$$

in its original oriented projection chart. At a terminal record its outgoing-next and incoming-previous shaded neighbors. Joining two actual identity-wire endpoints concatenates those two ordered sectors. To see the order directly, take a thin planar neighborhood of the joining arc: its two white sides join the complementary sectors, with no crossing or reflected child. Turns rotate the charts; the final closure uses (4.1). This constructs the full oriented ribbon graph, not merely its abstract incidence or component count.

All white face cycles are now determined. A facial walk entering along (u, v) leaves v along the neighbor preceding u in its cyclic order. The original shaded triples are some of these walks; the remaining walks are exactly the white regions. The prism maps (2.9) induce the same two pairings and the same sector concatenation. The mask proof shows that every other seam is flat white–white and every shaded edge remains. Thus the metric boundary has precisely this ribbon graph and every one of its colored facial cycles.

4.2 The reversible disk cut

Each original auxiliary circle lies in its own leaf ball, perpendicular to the projection sphere, with its recorded twice-punctured disk. The disks and their collars remain disjoint under physical turns and products. In either projection ball each auxiliary semicircle is boundary parallel by the corresponding half disk. Drilling the semicircle introduces one handle; cutting along that same bridge half disk cancels the handle in its explicit collar. The disjoint cancellations leave a ball. The main segments lie on the projection boundary; removing their neighborhoods creates ideal ends and no interior handle.

The cut ball has one ideal vertex for each paired main segment and one for each auxiliary semicircle. The exposed half disks give exactly the two shaded triangles above. The remaining regions have precisely the compiled white facial walks. Identify this labeled boundary sphere with ∂P_E and extend over the balls by radial extension after choosing interior points. Across the two balls restore the identically labeled white projection faces. For a pure leaf restore the shaded pairing fixing O_i and sending C_{i1} to C_{i0} and C_{i2} to C_{i3}. For an odd leaf use the original half-disk rotation, which pairs the corresponding triangles across copies. Reversing these cuts gives exactly Y_E^ε.

This is a degree-one topological identification with all components and paths retained. It does not interpret an intermediate gluing quadrilateral as the physical four-punctured boundary of a tangle: that assertion is unnecessary and generally false.

4.3 Complete returns and a common horosphere decoration

Take two oppositely oriented copies of P_E. For a pure leaf, an edge e of one shaded triangle has a distinct partner e' on the other. Its four occurrences are

$$e^{(0)},\ e'^{(0)},\ e'^{(1)},\ e^{(1)}.$$

The labeled shaded maps are inverse maps and the white maps identify equal labels across copies, so the return is pointwise identity on the edge. The angle sum is $4(\pi/2) = 2\pi$. An odd pairing changes the copy permutation in this return. It still has four distinct occurrences, since the two shaded triangles have no common edge and the two cut copies are distinct. The local pure and odd return identities proved in (8.2) below show that the full return is identity. Thus the quotient has neither an edge cone angle nor a residual translation along an edge.

At an ideal vertex all edges are right. Its small horosphere section is a finite Euclidean polygon with all angles $\pi/2$, hence a rectangle. Its side lengths are positive: the supporting planes are distinct, and coincident opposite sides would force the three-dimensional polyhedron into a plane. The colors alternate. Scale its horosphere until its two equal shaded side lengths are one. Each shaded ideal triangle now has its canonical three unit horocyclic segments, so every shaded isometry preserves these decorations, including the odd permutations. White maps preserve them because they identify the identical labeled copies. The rectangles therefore glue without dilation into smooth compact Euclidean tori, the original cusp tori of the cut. A common shrinking factor makes the horospheres disjoint; beyond them are complete horoball ends. The truncated core is compact. The resulting finite-volume metric is complete.

All cell attachments had disjoint interiors. There is one leaf pyramid and one prism per retained gate in each cut copy. Their volumes are πg and πm, respectively. Consequently $\mathrm{Vol}(Y_E^\varepsilon) = 2\,\mathrm{Vol}(P_E) = 2\pi F_0$ as an equality of ordinary metric volumes. This proves the metric part of Theorem 1.2, including the two-leaf strict case.

5 The primitive collar and its component information

The auxiliary apex retains its original leaf rectangle throughout the construction. In (2.6) its adjacent sides have lengths $2\sin\pi a$ and $2\cos\pi a$. The cusp has exactly two such rectangles. We trace the original disk paths in this unchanged collar and compare their lattice with its full two-rectangle area. This determines a primitive basis before considering a deformation.

Normalize the ordered leaf base to $(0,\infty,1,R)$. Let W_j be the white transition at edge j between projection charts, and let S, A, B be the pure and the two odd shaded maps, as defined in Section 7. The traced paths are

$$\lambda = W_2^{-1}W_0, \qquad \mu_0 = S^{-1}, \qquad \mu_- = A^{-1}W_0, \qquad \mu_+ = W_0^{-1}B. \tag{5.1}$$

The longitude follows the twice-punctured disk: it crosses white face 0 and returns through 2. The pure meridian crosses the shaded identification once in the reverse direction. The odd paths cross the respective half-disk pairing and white face 0: μ_- first crosses W_0 and then A^{-1}, while μ_+ first crosses B and then W_0^{-1}. These are the two original rotations through a half turn of the cut disk. They interchange the punctures and the two half disks, giving the same unmarked triangular pairing but meridians that differ by a full longitude. The retained negative crossing uses μ_- and the positive crossing uses μ_+ in the original disk frame. A physical turn transports the whole collar, not its sign.

At completeness the normalization from (2.6) is

$$M(z) = \frac{1+E}{2}\frac{z-1}{z-E}, \qquad R = \frac{(1+E)^2}{4E}, \qquad Z = M(\infty) = \frac{1+E}{2}.$$

For each word L in (5.1), substitute this R into (7.2), (8.1) and conjugate: $M^{-1}LM$ is proportional to $\left(\begin{smallmatrix}1 & t_L\\ 0 & 1\end{smallmatrix}\right)$. The four resulting translations are

$$\mu_0 = E-1, \qquad \lambda = -2(E+1), \qquad \mu_- = -2, \qquad \mu_+ = 2E. \tag{5.2}$$

For example S^{-1} may be represented by $\left(\begin{smallmatrix}-1-E & ER\\ -1 & 0\end{smallmatrix}\right)$, and λ by $\left(\begin{smallmatrix}1-ER & ER(E-1)\\ 1-E & E(E-R)\end{smallmatrix}\right)$. Their conjugations give respectively $E-1$ and $-2(E+1)$. Using $A^{-1}W_0$ and $W_0^{-1}B$ gives -2 and $2E$. The cross-copy words first return to the initial normalized chart, so this is one conjugation by M, not a mixture of the two projection frames. The actual torus area is $8\sin\pi a\cos\pi a = 4\sin 2\pi a$, and $|\operatorname{Im}(\overline{\mu_0}\lambda)| = 4\sin 2\pi a$. Thus the closed paths span the full lattice, not a proper sublattice. Replacing μ_0 by either odd meridian preserves this area. They too are primitive. As translation vectors on the corresponding half-twist partners,

$$\mu_\varepsilon = \mu_0 - \varepsilon\lambda/2.$$

This compares different marked tori; half a longitude is not being declared an integral curve on the pure torus. Since $(E+1)/(E-1) = -\mathrm{i}\cot\pi a$, the ratio λ/μ_ε is (1.6). This proves the remaining marking assertion of Theorem 1.2.

Proposition 5.1. *Every strict expression admits a choice of pure or odd leaves for which the main strands form one component. Either odd sign may be chosen, and subsequent full twists preserve the component count.*

Proof. Write the endpoint pairings as $I = (03)(12)$, $U = (01)(23)$ and $X = (02)(13)$. A leaf offers $\{U, X\}$, and a turn exchanges I, U and fixes X. Tracing the two actual product connections gives

	I	U	X
I	I	U	X
U	U	U and an internal circle	U
X	X	U	I

.

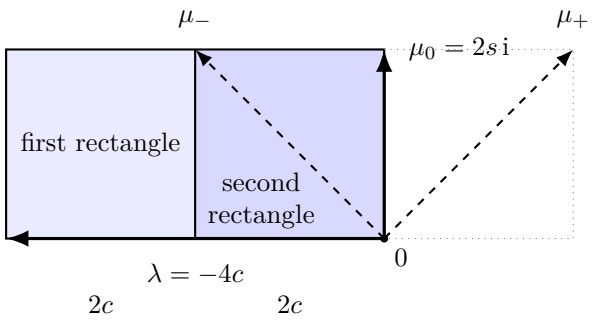


Figure 4: The original crossing-circle collar, rotated by $e^{-\pi \mathrm{i} a}$. Write $s = \sin \pi a$ and $c = \cos \pi a$. The pure meridian is $2s\,\mathrm{i}$, the disk longitude is $-4c$, and the two rectangles of width $2c$ and height $2s$ give the full fundamental area. The dashed meridians $\mu_- = \mu_0 + \lambda/2$ and $\mu_+ = \mu_0 - \lambda/2$ belong to their respective half-twist partners.

Discard only the product $U \cdot U$ in this existence argument. For sets $A = \{I, U\}$, $B = \{I, X\}$ and $C = \{U, X\}$, the output sets for AA, AB, AC, BB, BC, CC are respectively $A, \{I, U, X\}, C, B, \{I, U, X\}, A$, also in reversed orders. Thus every subexpression has at least two loop-free choices, starting with the leaf itself; the same is true if a child already offers all three. At least one choice at the root stack is U or X, both of which close to one component. Backtracking chooses the leaves. No identity of unmarked tangles or assertion about quantum connected-sum factors is used. □

6 A classical shear potential

Continue each occurrence of J holomorphically from its strict real argument. Its branch and additive constant are fixed by (1.2). For a leaf put

$$g(x,t) = J(2x) - 2J(x) - J(x-t) - J(x+t), \qquad v = g_t = \operatorname{Log} \frac{\sin \pi(x-t)}{\sin \pi(x+t)}. \tag{6.1}$$

At $t = 0$ this is $g(x)$ and $v = 0$. The local mixed pyramid has ordered base $(1, E/T, -1/T, -E)$, with $E = e^{2\pi \mathrm{i} x}$ and $T = e^{2\pi \mathrm{i} t}$. For small real t, its four half-arcs are $x-t, 1/2-x, x+t, 1/2-x$. The two positive tetrahedra across its first diagonal have angle triples $(x-t, 1/2-x, 1/2+t)$ and $(1/2-t, x+t, 1/2-x)$. The complementary Lobachevsky terms cancel, giving volume $\pi g(x,t)$. This fixes the local primitive without a quantum normalization argument.

For the retained sign ε, define the small hidden parameter $t = t(x, \chi)$ and its sheared potential by

$$\chi = t - \frac{\varepsilon v}{2\pi \mathrm{i}}, \qquad f_\varepsilon(x, \chi) = g(x,t) - \frac{\varepsilon v^2}{4\pi \mathrm{i}}. \tag{6.2}$$

At $(a, 0)$ the derivative of the first equation in t is $1 - \varepsilon \mathrm{i} \cot \pi a \neq 0$. The holomorphic implicit-function theorem applies. Differentiating both equations proves

$$\mathrm{d} f_\varepsilon = g_x \, \mathrm{d}x + v \, \mathrm{d}\chi. \tag{6.3}$$

The construction is separately even in each small χ because g is even in t and the shear equation is odd.

Choose independent real coordinates $\mathbf{y}$ on the original reduced wire space, keeping every turn and the complementary closing relation. The affine leaf and gate wires extend to its complexification. Set

$$\mathcal{A}_\varepsilon(\mathbf{y}, \boldsymbol{\chi}) = \sum_i f_{\varepsilon_i}(a_i(\mathbf{y}), \chi_i) + \sum_{\text{gates}} m. \tag{6.4}$$

At zero color this is exactly $\mathcal{F}$. Its nondegenerate internal Hessian is the imported strict variational theorem. A second application of the holomorphic implicit-function theorem therefore gives a unique small solution $\mathbf{y}_*(\boldsymbol{\chi})$ of $\partial_{\mathbf{y}}\mathcal{A}_\varepsilon = 0$ through the complete optimizer. Define

$$\Phi_\varepsilon(\boldsymbol{\chi}) = \mathcal{A}_\varepsilon(\mathbf{y}_*(\boldsymbol{\chi}), \boldsymbol{\chi}). \tag{6.5}$$

It is separately even, and the envelope rule gives $\partial_i \Phi_\varepsilon = v_i$. All these statements concern an explicitly defined classical germ, independently of any finite-level source.

Lemma 6.1. *With τ_i the actual primitive ratio in (1.6),*

$$\partial_i \partial_j \Phi_\varepsilon(0) = \pi \mathrm{i} \tau_i \, \delta_{ij}. \tag{6.6}$$

Proof. At a leaf, $g_{tt}(a, 0) = -2\pi c$ and $\chi_t = 1 - \varepsilon \mathrm{i} c$. Hence $f_{\chi\chi} = -2\pi c/(1 - \varepsilon \mathrm{i} c) = \pi \mathrm{i} \tau$. Also $g_{xt}(x, 0) = 0$. Every mixed wire–auxiliary derivative vanishes at zero, so eliminating the stationary wires adds no Schur-complement term to this block. Distinct leaves have no direct mixed auxiliary derivative in (6.4). □

7 Actual and auxiliary white transitions

An intermediate mask bit distinguishes two different roles for a matrix. At a zero bit it is the actual map on the adjacent white piece. At a one bit there is a shaded face instead; a *completion map* records the white transition that must be supplied from the other side when the edge becomes internal. It is not an extra face or a pairing assigned to that shaded triangle.

7.1 Universal local formulas

Normalize the four ordered face flags to $(0, \infty, 1, R)$. For the mixed leaf this is the explicit Möbius map

$$M_{x,t}(z) = \frac{1 + E}{1 + T} \, \frac{z - 1}{z - E/T}.$$

It sends $(1, E/T, -1/T)$ to $(0, \infty, 1)$. Applying it to the fourth corner and the apex gives

$$R = \frac{\cos^2 \pi x}{\cos^2 \pi t} = \frac{T(1 + E)^2}{E(1 + T)^2}, \qquad Z = \frac{1 + E}{1 + T}, \qquad Z^2 - (1 + E)Z + ER = 0. \tag{7.1}$$

In particular its ordered cross ratio is

$$R^{-1} - 1 = \frac{\sin \pi(x - t) \sin \pi(x + t)}{\cos^2 \pi x} = e^{-g_x}.$$

The sine quotient follows by subtracting the four phase coordinates; the derivative equality follows from (6.1). The first apex is Z and the second-copy apex is Z/E: the latter is the reflected minus-t chart. Reflection here means inversion of the phase coefficients, not complex conjugation of free parameters. Indeed the minus-t apex before reflection is $(1+E)/(1+T^{-1})$; replacing E, T by E^{-1}, T^{-1} gives Z/E. Every defining triple of face flags is distinct at completeness and remains so nearby. The two roots Z, TZ of the auxiliary fixed-point equation may coalesce at completeness; we require no distinctness of that pair there.

The four white or completion matrices are

$$\begin{aligned} W_0 &= \begin{pmatrix} 1 & 0 \\ 0 & E \end{pmatrix}, & W_1 &= \begin{pmatrix} 1 & E - 1 \\ 0 & E \end{pmatrix}, \\ W_2 &= \begin{pmatrix} E - R & R(1 - E) \\ E - 1 & 1 - ER \end{pmatrix}, & W_3 &= \begin{pmatrix} R & 0 \\ 1 - E & ER \end{pmatrix}. \end{aligned} \tag{7.2}$$

All matrices are projective. At a leaf, W_0, W_2 are its actual white maps: they fix the endpoints of the appropriate base edge and send Z to Z/E. The apex equation verifies the statement for W_2. The other two matrices are completion maps at its shaded cap bases.

We prove the whole prism identities in common world coordinates. Put $\alpha = e^{2\pi\mathrm{i}(z-x)}$, $\beta = e^{2\pi\mathrm{i}(x+y-z)}$, $\gamma = e^{2\pi\mathrm{i}(z-y)}$. The world prism is

$$\begin{aligned}(v_0, v_1, v_2, v_3) &= (1, \alpha, \alpha\beta, \alpha\beta\gamma),\\ P &= 1 + \alpha - \frac{\alpha[\alpha\beta(1+\gamma) - 1 - \alpha]}{\alpha^2\beta^2\gamma - \alpha}.\end{aligned} \tag{7.3}$$

For real strict parameters the two chord equations are

$$P + \alpha\overline{P} = 1 + \alpha, \qquad P + \alpha^2\beta^2\gamma\overline{P} = \alpha\beta(1+\gamma).$$

Eliminating $\overline{P}$ gives (7.3), so it is the same intersection as (2.8), not a second prism prescription. The formula now continues holomorphically in three independent phase variables. For an ordered quadruple (a, b, c, d) let $M(z) = (z-a)(c-b)/((z-b)(c-a))$ and $R = M(d)$, with the usual limits at infinity. Its world transition is $(M^{\#})^{-1}W_jM$, where $\#$ reciprocates the three phase variables. Use the ordered tuples (2.9), with wire phases $\beta\gamma, \alpha\beta, \alpha\beta\gamma$. Here is a full rational verification. Set $\Omega = \alpha\beta\gamma$ and $D = \alpha\beta^2\gamma - 1$. In this notation

$$P = \frac{\alpha\beta(\Omega + \beta\gamma - \gamma - 1)}{D}, \qquad \begin{array}{c|c} \text{face} & M(z)\\ \hline F & \dfrac{(z-1)(P-\alpha)}{(z-\alpha)(P-1)}\\ G & \dfrac{\alpha\beta - P}{z - P}\\ o & \dfrac{(z-1)(\alpha\beta-\alpha)}{(z-\alpha)(\alpha\beta-1)}\end{array}.$$

The fourth normalized vertex is always $M(d)$. Define the following five projective matrices:

$$\begin{aligned}K_0 &= \begin{pmatrix}\Omega - 1 & 1 - \alpha\Omega\\ \alpha(\beta\gamma - 1) & \alpha(1-\Omega)\end{pmatrix},\\ K_1 &= \begin{pmatrix}\Omega - 1 & \alpha\beta(1-\alpha\gamma)\\ \alpha D & \alpha^2\beta(1-\Omega)\end{pmatrix},\\ K_2 &= \begin{pmatrix}\Omega - 1 & \alpha\beta(1-\Omega\gamma)\\ \Omega(\alpha\beta - 1) & \alpha\beta\Omega(1-\Omega)\end{pmatrix},\\ K_3 &= \begin{pmatrix}-1 & \Omega + 1\\ 0 & \Omega\end{pmatrix}, \qquad K_* = \begin{pmatrix}-1 & P + \Omega P^{\#}\\ 0 & \Omega\end{pmatrix}.\end{aligned} \tag{7.4}$$

Substitute each displayed M and its fourth vertex into $(M^{\#})^{-1}W_jM$. Multiplication and cancellation of a scalar give the entire table

	$j=0$	$j=1$	$j=2$	$j=3$
F	K_0	K_1	K_*	K_3
G	K_*	K_1	K_2	K_3
o	K_0	K_1	K_2	K_3

. (7.5)

These cancellations are polynomial identities: insert the expression for P, clear D and the three normalization denominators, and compare the four matrix entries up to their common scalar. For example the shared-edge entry is the affine map $z \mapsto P^{\#} + (P - z)/\Omega$ from both

$F, 2$ and $G, 0$; the output cap entry $o, 3$ is $z \mapsto 1 + \Omega^{-1} - z/\Omega$. The displayed representatives specify every other entry without an unspecified map.

To identify the actual white triangles, K_1 sends $(P, \alpha, \alpha\beta)$ to $(P^{\#}, \alpha^{-1}, (\alpha\beta)^{-1})$, as substitution in its two linear rows verifies. Similarly K_3 sends $(\infty, 1, \Omega)$ to $(\infty, 1, \Omega^{-1})$. Thus they are the complete triangle maps, including translation. The table proves

$$\begin{aligned} W_{F,1} = W_{G,1} = W_{o,1} = W_{\text{top}}, \qquad W_{F,3} = W_{G,3} = W_{o,3} = W_{\text{bottom}}, \\ W_{F,2} = W_{G,0}, \quad W_{o,0} = W_{F,0}, \qquad W_{o,2} = W_{G,2}. \end{aligned} \tag{7.6}$$

with $W_{\text{top}} = K_1$ and $W_{\text{bottom}} = K_3$. All normalizations are invertible on the real strict chamber: for instance $D = e^{-2\pi i w} - 1 \neq 0$, $\beta\gamma = e^{2\pi i x} \neq 1$ and $\alpha\beta = e^{2\pi i y} \neq 1$. The geometric triples are distinct there. They remain so nearby. The rational identities hold off the real chamber as well and include a wire above $1/2$ without a completed-child assumption.

For turns and closure the corresponding identities are

$$\begin{aligned} T_Q(z) = \frac{R-1}{z-1}, \qquad C_Q(z) = \frac{z-R}{z-1}, \\ T_Q W_{j+1}(E, R) = W_j(E^{-1}, 1-R) T_Q, \\ C_Q W_j(E^{-1}, R) = W_{2-j}(E, R) C_Q. \end{aligned} \tag{7.7}$$

The first matrix sends the newly ordered first three corners to $0, \infty, 1$ after the cyclic shift; its fourth corner is $1 - R$. The second does the same for the full reversed closing order with the reciprocal wire phase. Inserting the four matrices (7.2) into either side gives equal projective matrices for each $j = 0, 1, 2, 3$. For example at $j = 0$ the first identity has matrices

$$\begin{pmatrix} 0 & E(R-1) \\ 1 & -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & R-1 \\ E^{-1} & -E^{-1} \end{pmatrix},$$

whose scalar ratio is E. The other entries follow by the same two-row multiplication; no endpoint derivative replaces the full map in (7.7).

7.2 The mask induction for the entire maps

We prove the following strengthened statement simultaneously through the original expression: at each output, zero bits carry actual white maps and one bits carry completion maps; at every internal shaded edge the actual adjacent white piece already has the recorded completion map. White pieces joined flatly carry the same entire map.

At a child cap seam the prism is white. If the child bit is zero, (7.6) identifies two actual maps on a flat seam. If it is one, the shaded triangle remains and the actual prism map supplies exactly its completion. At the common free edge the possible bit pairs are $(0,0), (1,0), (0,1)$. In the first case the actual white maps merge by $W_{F,2} = W_{G,0}$. In the other cases the unmasked child's actual map supplies the masked child's completion. The excluded pair $(1,1)$ would require HH. Finally the inherited output free maps and the two actual prism cap maps give precisely (3.3).

A turn transports this distinction by (7.7). At the closing face the strict complementary pair excludes two shaded bits on the same seam, as already proved. Zero–zero joins actual white pieces; one–zero supplies the required completion from the actual white side. No output face remains, so every completion map that enters a leaf edge return is now its actual neighboring white transition.

Stationarity of (6.4) matches the same complete complex cross ratios, by (6.3). Three vertices determine the Möbius map and the cross ratio matches the fourth. This proves the induction for the deformed charts as well. A deformed white polygon need not be planar; all its retained triangular pieces have one common transition by the proved identities.

8 The original group and the retained positive chain

Proposition 8.1. *The stationary family* (6.5) *determines a holomorphic framed representation of the original* $\pi_1(Y_E^\varepsilon)$ *through its complete holonomy. It retains a compatible positive ideal fundamental chain. All main peripheral groups remain parabolic, and the original auxiliary basis has exactly the logarithms* (1.7)*.*

We prove the group, chain and peripheral assertions in that order. In a leaf frame the pure shaded map and the two odd maps are

$$S = \begin{pmatrix} 0 & -ER \\ 1 & -1-E \end{pmatrix}, \quad A = \begin{pmatrix} R & -R \\ 1 & ER-1-E \end{pmatrix}, \quad B = \begin{pmatrix} R/E-1-1/E & R \\ -1 & R \end{pmatrix}. \tag{8.1}$$

They send the original ordered triples as follows: $S : (Z, \infty, 1) \mapsto (Z, 0, R)$, $A : (Z, \infty, 1) \mapsto (Z/E, R, 0)$, and $B : (Z, R, 0) \mapsto (Z/E, \infty, 1)$. With $S^\# = S(E^{-1}, R)$, direct multiplication gives

$$\begin{gathered} W_0 S = S^\# W_0, \quad W_2 S = S^\# W_2, \quad W_3 S = S^\# W_1, \\ W_0^{-1} B W_2^{-1} A = I, \quad W_2^{-1} B W_0^{-1} A = I, \quad W_1^{-1} B W_3^{-1} A = I. \end{gathered} \tag{8.2}$$

Equalities are in the projective group. In the displayed matrix representatives the scalar ratios in the first row are respectively E, E, ER. The three products in the second row are respectively $-R/E$, $-R/E$ and $(R-1)/E$ times the identity matrix. This proves the full identities by matrix multiplication for independent E, R; all these scalars are nonzero on the strict complete chamber and nearby. Reverse cycles give the inverse relations. The actual/completion-map induction is what permits their use on the enlarged syntax.

The original cut balls have a dual cell decomposition: paired faces give generators and edge cycles give the relations of its two-skeleton. Truncating the cusps gives the same group presentation; higher-dimensional cells add no fundamental-group relation. Every white seam and every pure or odd shaded return has now been verified by whole Möbius maps. Hence these maps define an actual holomorphic framed representation ρ_χ of $\pi_1(Y_E^\varepsilon)$ through its oriented complete holonomy. This conclusion uses the closed topological cut, not a cross-ratio argument for a hypothetical open four-punctured boundary.

There is also a compatible positive chain on the original manifold. At completeness first assemble the actual polyhedron P_E, whose ideal vertices are distinct except for the prescribed terminal pairs. Order these cut vertices and restrict the order to every constituent cell. Pulling triangulations agree on each shared quadrilateral. The order selects diagonals; orient every tetrahedron by the ambient oriented cell, not by its increasing vertex list. Coning from the first vertex to faces not containing it gives positive nondegenerate tetrahedra. Give the second projection copy the same labeled subdivision. White pairings match it, and shaded pairings are already triangular, so no order on the quotient cusp vertices is required.

Transport these finitely many flags by the actual rational charts. Their positive complete shapes remain in the positive half-plane on one small parameter polydisk. The face maps match the subdivided faces, and edge products stay exactly one. The continuous edge-log branches have their original sums there. Thus the representation has the same oriented ideal fundamental chain and boundary triangulation as the complete metric. We neither require deformed half-balls to be convex nor triangulate the filled manifold by new positive cells.

8.1 All main cusps remain parabolic

At corner C_j the actual small main meridian is $W_{j-1}^{-1} W_j$ or its inverse. One of these maps may have been supplied as a completion, but both are now actual white maps. Their derivatives at the corner agree: they are E^{-1} at C_0, C_2 and E at C_1, C_3, using a finite coordinate at infinity.

The meridian multiplier is identically one. The meridian is nonidentity at completeness and remains so nearby. It is therefore parabolic. Its commuting peripheral elements are translations in the same fixed-flag chart. This proves the statement for every main cusp, including any component closed inside an intermediate physical product.

8.2 The exact auxiliary logarithms

All the based curves (5.1) remain in the original leaf collar. Put

$$\rho = \frac{Z}{1+E-Z} = \frac{1}{T}, \qquad Q = \frac{E-Z}{Z-1} = \frac{ET-1}{E-T}. \tag{8.3}$$

In a frame sending the selected flag Z to infinity, the similarity multipliers of the four words are

$$\lambda : Q^2, \qquad \mu_0 : \rho, \qquad \mu_- : \rho Q, \qquad \mu_+ : \rho/Q. \tag{8.4}$$

For completeness, the following computation fixes every square and sign in (8.4). Replace each inverse in (5.1) by its adjugate; this changes only a projective scalar. The resulting representatives are

$$\begin{array}{c|c} \lambda & \begin{pmatrix} 1-ER & ER(E-1) \\ 1-E & E(E-R) \end{pmatrix} \\ \mu_0 & \begin{pmatrix} -E-1 & ER \\ -1 & 0 \end{pmatrix} \\ \mu_- & \begin{pmatrix} ER-E-1 & ER \\ -1 & ER \end{pmatrix} \\ \mu_+ & \begin{pmatrix} R-E-1 & ER \\ -1 & R \end{pmatrix} \end{array} .$$

For a matrix $(a\ b; c\ d)$ fixing Z, the derivative is $\det M/(cZ+d)^2$. Sending Z to infinity makes the affine similarity multiplier its reciprocal. With $Z = (1+E)/(1+T)$ and $R = T(1+E)^2/[E(1+T)^2]$, the needed determinants and denominator eigenvalues are

	$\det M$	$cZ+d$
λ	$E^2(R-1)^2$	$(ET-1)^2/(T+1)^2$
μ_0	ER	$-(E+1)/(T+1)$
μ_-	$E^2R(R-1)$	$(E+1)(ET-1)/(T+1)^2$
μ_+	$R(R-1)$	$-(E+1)(E-T)/[E(T+1)^2]$

. (8.5)

The quotient $(cZ+d)^2/\det M$ in each row is exactly the corresponding factor in (8.4). Since $\rho = e^{-2\pi i t}$ and $Q = e^{-v}$, their small logarithms are

$$U = -2\pi i t + \varepsilon v = -2\pi i \chi, \qquad V = -2v. \tag{8.6}$$

These are similarity logs, twice selected eigenvalue logs. Their constants are zero at completeness. The envelope rule proves (1.7); physical turns only conjugate the words.

If $\chi_i = 0$, separate evenness gives $V_i = 0$ as well. The auxiliary meridian remains nonidentity nearby, so an auxiliary cusp left unfilled also remains a genuine parabolic translation group. All flags and defining face triples stay in a finite collection of bounded local frames. This proves Proposition 8.1.

9 Independent signed slopes and smooth completion

Fix the subset I and set all other χ_i equal to zero. For $i \in I$ write $r_i = 1/p_i$. Solve

$$V_i(\boldsymbol{\chi}) + r_i U_i(\boldsymbol{\chi}) = -2\pi \mathrm{i} r_i. \tag{9.1}$$

At $\mathbf{r} = 0$ the solution is $\boldsymbol{\chi} = 0$. Equations (1.7) and (6.6) give the invertible Jacobian

$$D_{\boldsymbol{\chi}} V(0) = -2\pi \mathrm{i}\, \mathrm{diag}(\tau_i)_{i \in I}. \tag{9.2}$$

The implicit-function theorem gives a unique solution on one reciprocal-slope polydisk. Shrink it until the representation and positive original chain are in their common fixed neighborhood. For real signed p_i this imposes a lower bound on each $|p_i|$, with no restriction on their ratios. For $p_i \neq 0$ the equations are

$$U_i + p_i V_i = -2\pi \mathrm{i}. \tag{9.3}$$

They are precisely the critical equations of (1.8).

The peripheral direction is actually loxodromic at these solutions. Separate evenness makes the apparent quotient $\widetilde{\tau}_i = V_i/U_i$ removable, holomorphic near zero, and equal to τ_i there. Shrink once more so that $\operatorname{Im} \widetilde{\tau}_i > 0$. Equation (9.1) yields

$$\chi_{i,*} = \frac{r_i}{\widetilde{\tau}_i + r_i}, \qquad \operatorname{Re} V_i = \frac{2\pi r_i^2 \operatorname{Im} \widetilde{\tau}_i}{|\widetilde{\tau}_i + r_i|^2} > 0. \tag{9.4}$$

This is valid for either sign of a nonzero real reciprocal slope. It is a classical consequence of the positive primitive cusp ratio.

We now apply the precise near-complete completion result [4, Proposition 6.1]. Its input is an oriented complete finite-volume manifold with marked primitive cusp bases, and an actual nearby framed representation with small similarity logs in bounded frames. At unfilled cusps both multipliers must be one. At each filled cusp a primitive integral combination of the small logs must equal $2\pi\mathrm{i}$ or $-2\pi\mathrm{i}$. It then completes the original topological filling to a smooth complete hyperbolic metric.

Every hypothesis has been supplied on this enlarged syntax: Theorem 1.2 constructs the actual complete exterior; the unchanged two-rectangle collar proves its primitive basis; Sections 7–8 construct the actual framed representation and the original group relations; all unfilled cusps have translation holonomy; and (9.3) uses the primitive vector $(1, p_i)$. The fixed small parameter neighborhood is chosen before the slopes.

9.1 Why the integer equation gives one actual angular turn

We recall the part of the imported proof that prevents an exponential period equation from being mistaken for completion. On a complete cusp link use a translation lattice $\Lambda \subset \mathbb{C}$ and its fixed generator paths. A nearby peripheral action is affine, $A_\lambda(z) = a_\lambda z + b_\lambda$, in a bounded moving flag frame. A smooth partition η on the complete plane with $\sum_{\lambda \in \Lambda} \eta(w - \lambda) = 1$ gives

$$d_\rho(w) = \sum_{\lambda \in \Lambda} \eta(w - \lambda) A_\lambda(w - \lambda).$$

On a fundamental parallelogram this is C^1-close to the complete identity and has positive real Jacobian; equivariance propagates this fact. With the real peripheral logarithmic character in the height coordinate it gives a nonsingular developed cusp, matched over a compact collar to a nearby developing map on the original truncated core.

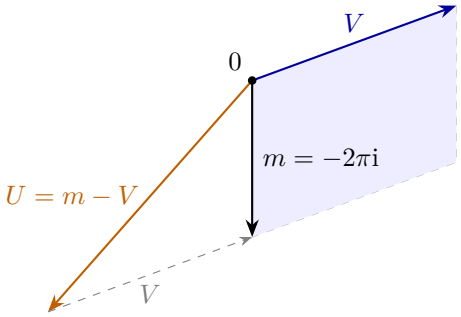
At a filled cusp a multiplier has nonunit modulus by (9.4). Commutativity gives a second common fixed point q besides infinity. Since the original complete translations are nonzero,

$|q| \to \infty$ near completeness. The developed fixed generator paths stay bounded, with bounded length. They miss q, and their increments of $f = \operatorname{Log}(d_\rho - q)$ tend to zero. Quantitatively, a path of length at most L contained in $|d_\rho| \le R$ has $|\Delta f| \le L/(|q| - R)$. Thus those increments are the prescribed small logs U, V, without an added integral multiple of $2\pi\mathrm{i}$, *before* multiplication by the large integer p.

The metric $|df|^2$ descends to a compact nonsingular flat torus. Its complete lifted developing map is a diffeomorphism to $\mathbb{C}$, so U, V form its actual rank-two period lattice. The developed end contains a punctured metric tube with cylindrical metric

$$dR^2 + \cosh^2 R\, dX^2 + \sinh^2 R\, d\theta^2. \tag{9.5}$$

A complex period L acts by $(X, \theta) \mapsto (X + \operatorname{Re} L, \theta + \operatorname{Im} L)$. The original primitive slope has period $U + pV = -2\pi\mathrm{i}$, exactly one angular turn. Its complementary primitive period V has nonzero real part. Adding $R = 0$ produces the smooth solid torus with this original meridian. The slope fixes both q, ∞ and has multiplier one, so it is the identity in the group, not a nontrivial unipotent with the same eigenvalues. The remaining ends are full horoball quotients, and the middle region is compact. Completeness and finite volume follow.



Original primitive basis: U, V.

Filled meridian: $m = U + pV$.

The fixed small increments are established before forming m.

Figure 5: The developed logarithmic link of a filled original crossing circle. The old basis is U, V, while $m = U + pV = -2\pi\mathrm{i}$ is the primitive meridian of the added solid torus. The picture shows $p = 1$; the proof uses the small periods before multiplying by any integer p. In the tube metric, m makes one angular turn and V gives a nonzero axial translation.

This verifies the metric conclusion of Theorem 1.3 for the actual disk-filled exterior. In particular high-index estimates for a quantum coefficient, source positivity and whole character-variety rigidity have played no role.

10 Absolute volume on the original chain

Set $G = 2\pi\mathrm{i}\Phi_\varepsilon$. The exact peripheral identity gives

$$\mathrm{d}G = \frac{1}{2}\sum_i V_i\, \mathrm{d}U_i, \qquad \operatorname{Im} G(0) = 2\pi F_0 = \operatorname{Vol}(Y_E^\varepsilon). \tag{10.1}$$

This is the $V\,dU$ polarization. A primitive of $-\frac{1}{2}\sum U\,dV$ differs by the boundary term $\frac{1}{2}\sum UV$; its regulator correction cannot be reused unchanged here.

Let $\mathcal{V}(\rho)$ be the signed sum of the ideal tetrahedron volumes of the retained original chain. The local boundary differential [4, equation (7.3)] reads, for similarity logs,

$$\mathrm{d}\mathcal{V} = \frac{1}{2}\sum_i \bigl(\operatorname{Re} V_i\, \mathrm{d}\operatorname{Im} U_i - \operatorname{Re} U_i\, \mathrm{d}\operatorname{Im} V_i\bigr). \tag{10.2}$$

Its hypotheses hold on our positive nondegenerate chain: internal faces cancel by the actual maps, edge-log sums stay fixed, and the primitive cusp orientation is $\operatorname{Im}(\lambda/\mu) > 0$ by (1.6). All main logs vanish. The proof of that local identity differentiates Bloch–Wigner volume and cancels normal-face flows; it uses the positive ordered model $(0, \infty, 1, z)$ and the primitive cusp intersection pairing. It is not a theorem restricted to the complete packing of a particular diagram.

Taking the imaginary part of (10.1) and differentiating its boundary correction now gives

$$\mathcal{V}(\rho_{\boldsymbol{\chi}}) = 2\pi \operatorname{Re} \Phi_{\varepsilon}(\boldsymbol{\chi}) - \frac{1}{2} \sum_{i} \operatorname{Re} U_i \operatorname{Im} V_i. \tag{10.3}$$

The integration constant is fixed by the actual degree-one complete cell volumes, not just by a derivative computation. At a filling solution write $\chi_i = u_i + \mathrm{i}v_i$. Since $V_i = 2\pi\mathrm{i}(\chi_i - 1)/p_i$,

$$-\frac{1}{2} \operatorname{Re} U_i \operatorname{Im} V_i = -\frac{2\pi^2}{p_i} v_i(u_i - 1) = 2\pi \operatorname{Re}\left[\frac{\pi\mathrm{i}}{2p_i}(\chi_i - 1)^2\right]. \tag{10.4}$$

Unfilled auxiliaries have $U_i = V_i = 0$.

10.1 The same boundary triangulation after filling

It remains to identify $\mathcal{V}$ with the volume of the smooth metric just constructed. We verify the chain hypotheses of [4, Section 7.3]. At a filled cusp conjugate the common fixed points to $0, \infty$. On each triangle of the retained boundary triangulation the ideal conical development is $(a(w), Yb(w))$, with $b > 0$. All functions and their derivatives are bounded on a finite fundamental set. Choose an equivariant axis map $(0, Ye^{\eta(w)})$, where η has the real peripheral logarithmic character. On the *same* triangulation set

$$H_s(w, Y) = \left(sa(w), Y \exp\big((1-s)\eta(w) + s \log b(w)\big)\right), \qquad 0 \le s \le 1. \tag{10.5}$$

Both horizontal coordinates and heights have the required equivariance, and the homotopy agrees along every face. Its height is comparable to Y, horizontal derivatives are bounded, and height derivatives are $O(Y)$. The volume form $y^{-3}\, dx_1 \wedge dx_2 \wedge dy$ therefore bounds its three-dimensional boundary sweep by $O(Y^{-2})$.

Once the primitive slope is killed, extend that axis-valued boundary over the added solid torus. The remaining cyclic group acts along the axis, so all added vertices can lie on the axis and all added tetrahedra have zero signed volume. At unfilled cusps the same vanishing boundary estimate applies. Stokes on finite truncations, followed by $Y \to \infty$, identifies the retained ideal chain with the filled equivariant fundamental class. Comparing it with the complete metric developing map on a common subdivision gives

$$\mathcal{V}(\rho_{\boldsymbol{\chi}_*}) = \operatorname{Vol}(Y_E^{\varepsilon}(\mathbf{p}; I)).$$

An unrelated chain with the same peripheral differential would not suffice; (10.5) supplies the absolute comparison. Combining it with (10.3) and (10.4) proves (1.9).

11 Proof boundary and finite algebra

The strict variational existence, Hessian and boundary-profile identities are imported from [1], with their precise nonmetric content in Section 2. The only local volume input is the classical ideal-tetrahedron formula (2.5). The local cells, all supporting planes and cross ratios, entire rational prism maps, collar words and similarity multipliers are proved here. The individual-mask induction establishes their enlarged closed topology and every actual

versus completion-map status. The earlier completed-root papers [2, 3] are archival accounts of that narrower construction, not prerequisites of the present local calculations or global theorem. Smooth metric completion and the local volume differential are the explicitly stated inputs from [4], with all hypotheses verified here.

No source asymptotic, full spectral bound or nonzero Fourier contribution is claimed. To identify the classical action with a quantum packet one must independently match each endpoint-normalized leaf action, including its color-dependent additive constant, and then eliminate these same wires. A further whole-source estimate and filling noncancellation theorem are needed for an invariant volume limit. The geometry alone does not supply them. Similarly no equivalence with a completed-root presentation is used, and no moving main-meridian or preferred-longitude deformation claim is made.

The accompanying `check.py` verifies the finite mask closure and its allowed edge pairings, the literal endpoint-connection table, projective turn/closing and pure/odd return identities, the independent-phase prism table, complete collar conjugations, deformed peripheral multipliers, and the primitive Hessian and critical boundary normalization. It includes altered mask, pairing and factor-of-two controls. It writes a source-hashed JSON record with `-output`. These exact finite checks do not prove convexity, the variational existence theorem, the original manifold identification, implicit-function theory, smooth completeness or the chain comparison.

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